

Lesson plan

Name of the Programme:	B.Tech(Hons.)	
Title of the Course:	Database Management System (DBMS)	
Course code	CS1211	
Semester:	II	
Names of the Course Instructor(s)	Prof. Sonam V Maju, Prof. Sheba Pari, Prof. Aparna R, Prof.Evlin Vidyulatha, Prof.Jeethu, Prof.Vinod, Prof. Sunil Kumar J	
Course credit	No. of hours per week (2:1:2)	Total no. of Teaching hours
04	05 Hours	75 Hours

Session	Module	Topics	Pedagogy	Textbook Reference
1	1	Introduction to DBMS	Interactive PPT	Silberschatz Ch.1
2	1	File System vs DBMS	Interactive PPT	Navathe Ch.1
3	1	DBMS Architecture	Interactive PPT	Silberschatz Ch.2
4	1	Data Models, Schema	Interactive PPT	Navathe Ch.2
5	1	DB Languages : DML, DDL	Interactive PPT	Silberschatz Ch.4
6		DB Languages: DCL, TCL	Interactive PPT	
7		DB Languages	Interactive PPT	

8	1	Metadata, Storage	Interactive PPT	Silberschatz Ch.13
9	1	Backup & Recovery	Interactive PPT	Silberschatz Ch.18
10	1	Practice Session – Module 1	Problem Solving	Combined
11	1	Tutorial: DB Architecture	Tutorial	Silberschatz Ch.2
12	1	Lab 1 – DDL Commands	Hands-on SQL	Silberschatz Ch.4
13				
14	1	Lab 2 – DML Commands	Hands-on SQL	Silberschatz Ch.4
15				
16	1	Lab 3 – SQL Functions	Hands-on	Navathe Ch.5
17				
18	2	ER Concepts	Interactive PPT	Navathe Ch.7
19	2	ER Concepts	Interactive PPT	Navathe Ch.7
20	2	ER Notations	Interactive PPT	Navathe Ch.7
21	2	Weak Entities	Interactive PPT	Navathe Ch.7
22	2	ER → Tables	Interactive PPT	Navathe Ch.9

23	2	ER → Tables	Interactive PPT	Navathe Ch.9
24	2	Keys & Constraints	Interactive PPT	Silberschatz Ch.3
25	2	Relational Algebra	Interactive PPT	Silberschatz Ch.6
26	2	Relational Algebra	Interactive PPT	Silberschatz Ch.6
27	2	Relational Calculus	Interactive PPT	Silberschatz Ch.6
28	2	Relational Calculus	Interactive PPT	Silberschatz Ch.6
29	2	Practice – ER & RA	Worksheets	Combined
30	2	Tutorial: ER Diagram Practice	Hands-on	Navathe Ch.7
31	2	Tutorial: Relational Algebra	Worksheet	Silberschatz Ch.6
32	CIE-1	Internal Exam – CIE 1 (2 Hours)	Written Test	Modules 1 & 2
33	3	SQL Basics	Interactive PPT	Silberschatz Ch.8
34	3	Tables & Indexes	Interactive PPT	Navathe Ch.5
35	3	Queries & Subqueries	Interactive PPT	Silberschatz Ch.8

36	3	Joins	Interactive PPT	Silberschatz Ch.8
37	3	Joins	Interactive PPT	Silberschatz Ch.8
38	3	Views & Triggers	Interactive PPT	Navathe Ch.5
39	3	SQL Practice Session	Hands-on	Combined
40	Lab	Lab 4 – Operators	Hands-on	Navathe Ch.5
41				
42	Lab	Lab 5 – Joins	Hands-on	Silberschatz Ch.8
43				
44	Lab	Lab 6 – Grouping & Ordering	Hands-on	Silberschatz Ch.8
45				
46	3	Tutorial: SQL Query Practice	Worksheet	Silberschatz Ch.8
47	4	Functional Dependencies	Interactive PPT	Navathe Ch.10
48	4	1NF	Interactive PPT	Navathe Ch.10
49	4	2NF	Interactive PPT	Navathe Ch.10
50	4	3NF	Interactive PPT	Navathe Ch.10

51	4	BCNF	Interactive PPT	Navathe Ch.10
52	4	Lossless Join	Interactive PPT	Silberschatz Ch.8
53	4	Dependency Preservation	Interactive PPT	Silberschatz Ch.8
54	4	Normalization Worksheets	Practice	Combined
55	4	Normalization Case Study	Activity	Combined
56	4	Tutorial: Normal Forms (All NFs)	Worksheet	Navathe Ch.10
57	CIE-3	Internal Exam – CIE 3 (2 Hours)	Written Test	Modules 3 & 4
58	5	ACID Properties	Interactive PPT	Silberschatz Ch.14
59	5	Serializability	Interactive PPT	Silberschatz Ch.14
60	5	Serializability	Interactive PPT	Silberschatz Ch.14
61	5	Conflict Serializability	Interactive PPT	Silberschatz Ch.14
62	5	Conflict Serializability	Interactive PPT	Silberschatz Ch.14
63	5	View Serializability	Interactive PPT	Silberschatz Ch.14

64	5	View Serializability	Interactive PPT	Silberschatz Ch.14
65	5	Deadlocks	Interactive PPT	Silberschatz Ch.17
66	5	Locking Protocols	Interactive PPT	Silberschatz Ch.17
67	5	Timestamp Protocols	Interactive PPT	Silberschatz Ch.17
68	5	MVCC	Interactive PPT	Silberschatz Ch.17
69	5	Validation Techniques	Interactive PPT	Silberschatz Ch.17
70	5	Practice – Transactions	Problem Solving	Combined
71	5	Tutorial: Transaction Problems	Worksheet	Silberschatz Ch.14
72	Lab	Lab 7 – Backup & Recovery	Demo	Silberschatz Ch.18
73				
74	Lab	Lab 8 – Users & Privileges	Hands-on	Combined
75				

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Quiz Graded Component 1 (based on Theory + SQL Queries) - Assignment on ER Diagrams	CIE :15 marks out of 70 Assignment: 5 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Lab & viva - Lab Examination: CodeTantra - Viva Voce	10 marks out of 70 10 marks out of 70 5 marks out of 70

Semester End Exams (SEE): 30% (Refer Table 4)	
Mode of Exam	Jury
Weightage of exam	30%

Course Outcomes- Programme Outcomes Matrix

Program Outcome Course Outcome		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	To describe fundamental concepts of database systems, including data models, architectures, and schema design.	3	2	-	2	-	-	-	-	-	2	-	-
CO2	To analyse and decompose database designs using normalisation and functional dependencies to eliminate redundancy and improve efficiency	3	3	-	3	-	-	2	2	-	2	-	-

CO3	To solve real-world problems by applying the Database Management System concept.	2	3	2				2	2		3		
CO4	Learn the necessary skills for transaction processing and effective database design.	3	2	2	3			2	2		1		

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Quiz Graded Component 1	Quiz will be conducted for 15 marks. Each right answer carries 0.5 mark each. No negative marking for wrong answers.
	Assignment Component-2	Entity–Relationship (ER) Diagrams are essential tools in database design. They help identify entities, attributes, and relationships required to model real-world systems. This assignment will assess the student’s understanding of conceptual database design using ER modeling techniques. Students must choose a real-life scenario, identify all

		required components, and design an ER Diagram adhering to standard notations. <i>(Refer Table-1)</i>
--	--	---

Table-1:

Graded Component-2 (5 Marks):

Assess students' ability to complete ER Diagram,

Criteria	Excellent (5 Marks)	Satisfactory (3 Marks)	Needs Improvement (0 Marks)
Completion of Courses (5 Marks)	ER diagram is accurate, complete, neatly presented, uses correct ER notations, includes correct entities, attributes (PK/FK), relationships, and cardinalities.	The diagram is mostly correct but may have minor errors in entities, relationships, or notation; presentation is acceptable.	Diagram is incomplete, incorrect, missing key entities/relationships, or not following ER notation.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Practical Component	Lab + Viva = 10 marks <i>(Refer Table-2)</i> CodeTantra Practicals : 10 Marks <i>(Refer Table-3)</i> Viva :5 Marks <i>(Refer Table-3)</i>

Table 2:

Lab Execution rubrics (Max: 6 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)
1.	Understanding of problem statements and preparedness (2 Marks)	Completeness of lab setup (e.g., MySQL, tools pre-installed)	Students exhibit a thorough understanding of the requirements and fully prepared with all tools and setup ready	Students have sufficient understanding of requirements and are partially prepared with installation.	Students exhibit little understanding of requirements and Poorly prepared (multiple missing setups)
2.	Execution (2 Marks)	Running commands, executing tasks, capturing results	Executes all steps correctly and efficiently	Executes with minor issues or guidance	Unable to execute key tasks independently
3.	Results and Documentation (2 Marks)	Screenshots, logs, steps, conclusions in lab report	Well-documented with clear evidence and interpretation	Incomplete or unclear documentation	Minimal documentation, hard to follow
Viva Voce Rubrics (Max: 4 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)
1.	Conceptual Understanding (2 Marks)	Viva Voce	Thoroughly explains the related concepts	Adequately explains the related concepts	Unable to explain related concepts
2.	Problem solving ability (2 Mark)	Viva Voce	Demonstrates clear, logical approach to solving problems, with reasoning	Identifies issues and partially solves problems	Struggles to solve problems

Table-3

Sl. No	Criteria (Marks)	Measuring Method	Excellent	Good	Poor
1	Concept Clarity & Problem Understanding (3 Marks)	Student explains task purpose, logic, and expected output	Clearly explains the problem, shows strong conceptual understanding, identifies correct approach (3 Marks)	Basic understanding with minor gaps in explanation or reasoning (2 Marks)	Unable to clearly explain the problem; weak conceptual understanding (1 Mark)
2	Implementation & Execution Quality (5 Marks)	Accuracy of commands/code, order of steps, correctness of output	Executes all steps correctly, follows proper procedure, produces accurate output independently (5 Marks)	Executes most steps with minor errors or occasional guidance (3–4 Marks)	Multiple errors, incomplete execution, or heavy dependence on help (1–2 Marks)
3	Output Validation & Presentation (2 Marks)	Clarity, correctness, organization, and explanation of results	Output fully correct, clearly organized, well-explained (2 Marks)	Output mostly correct but lacks clarity or proper formatting (1.5 Marks)	Output incorrect, unclear, or poorly organized (1 Mark)

Viva Voce Rubrics (Max: 5 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (5 Marks)	Good (3 Mark)	Poor (1.5 Marks)
3.	Conceptual Understanding (2 Marks)	Viva Voce	Thoroughly explains the related concepts	Adequately explains the related concepts	Unable to explain related concepts
4.	Problem solving ability (3 Mark)	Viva Voce	Demonstrates clear, logical approach to solving problems, with reasoning	Identifies issues and partially solves problems	Struggles to solve problems

Table- 4:

Grading Rubrics for Project (Total: 30 Marks)					
Sl. No	Criteria (Marks)	Excellent	Very Good	Good	Needs Improvement
1	Problem Relevance, Background & Context (6 Marks)	Problem statement is highly relevant to DBMS; provides clear context, motivation, scope, data description, and justification for database solution. (6 M)	Problem is relevant; background is clear with minor gaps in context or justification. (4–5 M)	Covers essential elements but lacks depth, clarity, or strong connection to DBMS concepts. (2–3 M)	Problem relevance is weak or poorly justified; background vague or incomplete. (0–1 M)

2	Database Design & Modeling (ERD, Schema, Normalization) (8 Marks)	ER diagram, schema, constraints, and relationships are well-designed; normalization up to BCNF or 3NF thoroughly applied; design shows strong logical correctness. (8 M)	Design is correct with minor issues; normalization applied up to at least 2NF/3NF with small gaps. (6–7 M)	Basic design present but with noticeable errors in attributes, keys, or normalization steps. (3–5 M)	Design incomplete, incorrect, or lacks normalization and clarity. (0–2 M)
3	Implementation: SQL Queries, Transactions & Execution (8 Marks)	SQL queries well-implemented (DDL, DML, joins, views, triggers, transactions); outputs accurate; project demonstrates strong technical mastery. (8 M)	Most queries correctly implemented with minor errors; outputs largely correct. (6–7 M)	Basic queries working but missing key components (joins, constraints, triggers, or transactions). (3–5 M)	Major errors in SQL implementation ; incomplete execution; missing essential DBMS components. (0–2 M)
4	Insight, Analysis & Originality (4 Marks)	Demonstrates strong analytical thinking, unique approach, performance evaluation, or meaningful	Provides good analysis and useful insights with minor gaps; some originality present. (3 M)	Basic analysis included but lacks depth, critical thinking, or creativity. (2 M)	Minimal insight; little to no originality or analysis. (0–1 M)

		insights; shows originality. (4 M)			
5	Project Demonstration & Viva Voce (3 Marks)	Presents clearly and confidently; demonstrates the system smoothly; answers all questions accurately with strong conceptual clarity. (3 M)	Presentation is clear with minor gaps; handles questions reasonably well. (2 M)	Presentation understandabl e but unstructured; difficulty answering technical questions. (1 M)	Presentation unclear; unable to explain project or answer questions. (0 M)
6	Project Report (1 Mark)	Well-structure d, complete, follows template, includes all required sections, and submitted on time. (1 M)	Mostly complete, organized, and follows template with small omissions. (0.5 M)	Some sections missing or unclear; partial template adherence; delayed submission. (0–0.5 M)	Poorly organized report; major sections missing; significantly delayed submission. (0 M)